Exp No:13

Case Study on the Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Aim

To study the limitations of low-code platforms using a complex enterprise use case and understand the challenges involved in scalability, customization, integration, security, and performance.

Record Status

Case study completed based on the analysis of a complex enterprise application scenario.

Concept

Low-code platforms allow applications to be developed using visual tools, predefined components, and minimal hand-written code. Zoho Creator is an example of a low-code application development platform.

Although low-code platforms reduce development time and simplify application creation, they may have limitations when handling complex enterprise requirements, advanced business logic, large data volumes, specialized integrations, and strict security requirements.

Procedure

1. Select a complex enterprise use case, such as a Loan Application Management System for a large financial organization.
2. Identify the application's requirements, including applicant registration, credit assessment, document processing, approval workflows, and reporting.
3. Examine how a low-code platform such as Zoho Creator can implement these requirements.
4. Analyze customization options for complex business rules and workflows.
5. Evaluate integration requirements for external banking systems, credit bureaus, and document verification services.
6. Study performance and scalability challenges when handling large numbers of users and records.
7. Examine security, access control, and data governance requirements.
8. Identify the situations in which custom programming or external services may be necessary.
9. Summarize the limitations and recommend suitable solutions.

Application Design

Case Study: Enterprise Loan Processing System

A financial organization wants to automate loan applications for thousands of customers. The system must collect applicant information, verify documents, assess creditworthiness, route applications for approval, and generate reports.

Proposed low-code solution using Zoho Creator:

- Forms collect applicant and loan details.
- Reports display application records and processing status.
- Workflows automate routine approval steps.
- Deluge scripts implement business rules and calculations.
- Role-based permissions restrict access to sensitive information.
- Integrations connect the application to approved external services where supported.

Limitations identified

1. **Complex customization:** Highly specialized algorithms or business rules may require additional code or external services.
2. **Scalability:** Performance must be evaluated for large record volumes, concurrent users, and complex workflows.
3. **Integration:** External banking systems may require APIs, authentication, custom connections, or middleware.
4. **Platform dependency:** Applications may rely on platform-specific features and scripting, making migration more difficult.
5. **Security and compliance:** Enterprise deployments require careful configuration of access controls, audit requirements, and data protection.
6. **Cost and usage limits:** Subscription plans, API limits, and platform resource limits may affect large-scale deployments.
7. **Limited control:** Developers may have less control over the underlying infrastructure and runtime compared with fully custom-built systems.

Observations

1. Low-code platforms accelerate the development of standard business applications.
2. Visual development tools simplify forms, reports, and routine workflows.
3. Advanced enterprise requirements may exceed built-in capabilities.
4. Integration and performance depend on platform features, configuration, and external services.
5. A hybrid approach combining low-code applications with custom services can address some limitations.

Validation and Discussion

The case study shows that a low-code platform can support many functions of an enterprise loan processing system. However, complex credit-risk models, high-volume integrations, specialized compliance controls, and demanding performance requirements require additional evaluation.

Before choosing a platform, organizations should test representative workloads, verify integration capabilities, review security requirements, and estimate subscription and maintenance costs.

Not every limitation requires abandoning low-code development. Appropriate architecture, optimized workflows, external APIs, and custom services can extend the platform where necessary.

Result

Thus, the limitations of low-code platforms were analyzed through an enterprise loan processing case study. The study concludes that low-

code platforms are effective for rapid application development, but complex enterprise systems may require custom code, external integrations, and careful performance and security planning.